

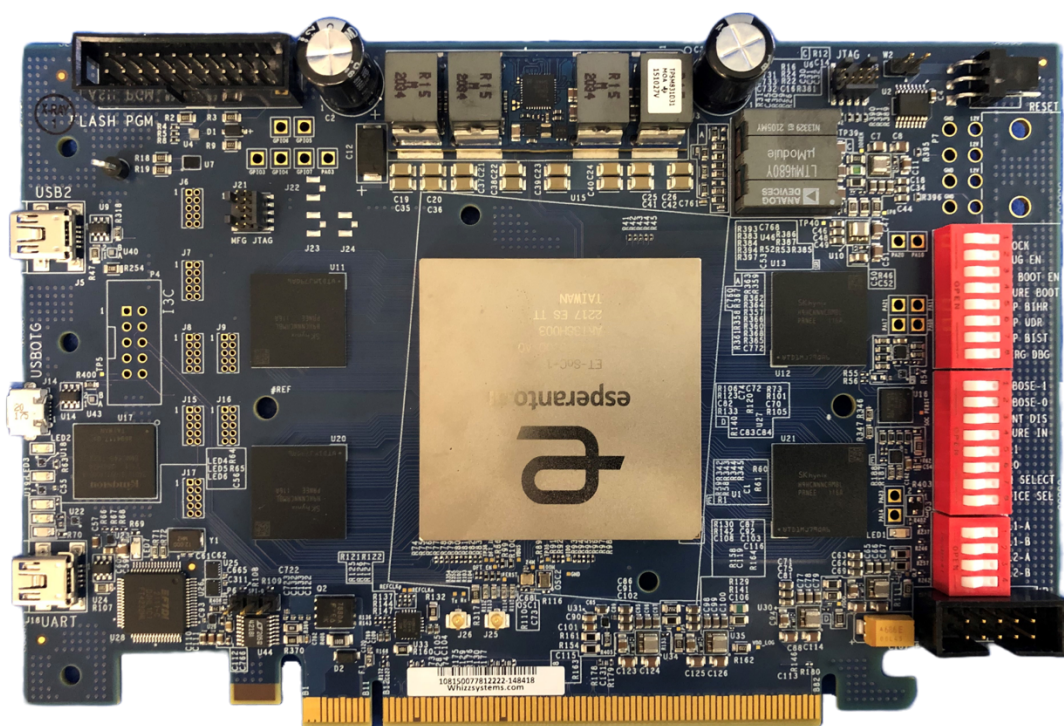
PCIe Dev Card (V3)

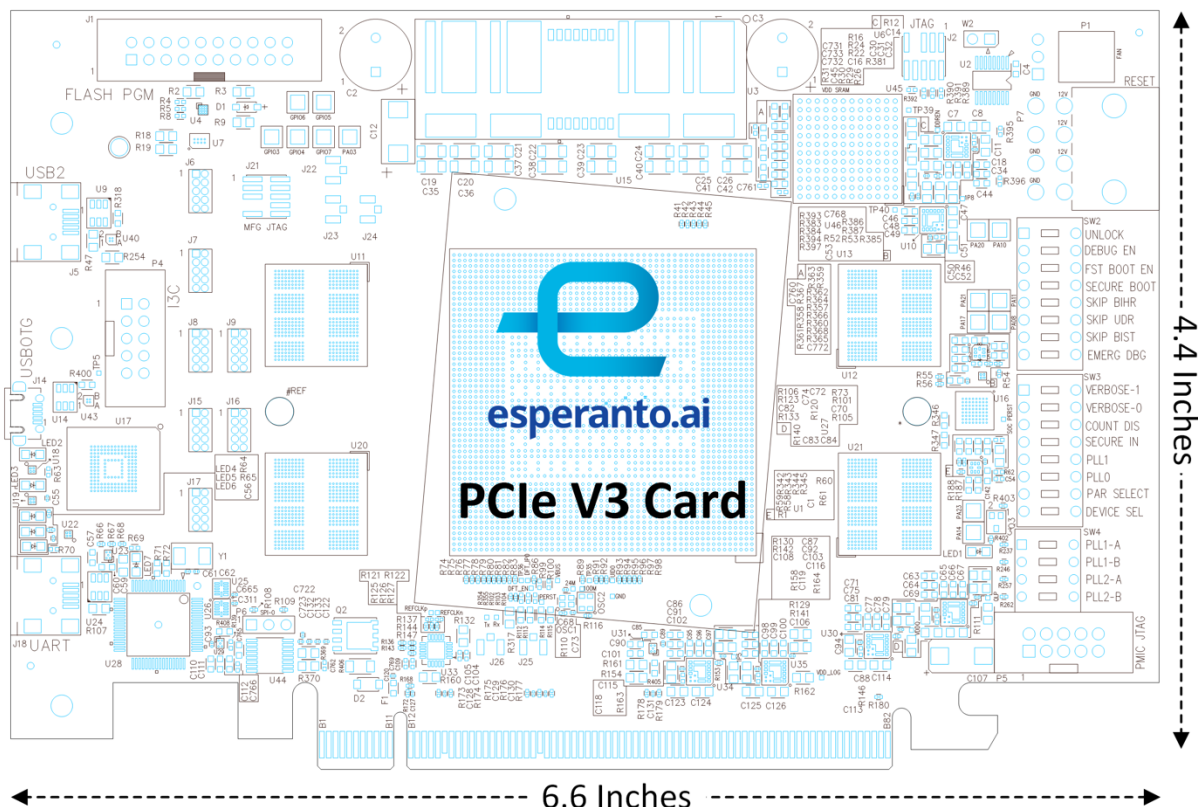
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- [PCIE_5xx_OPS_Sheet](#)
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General Information



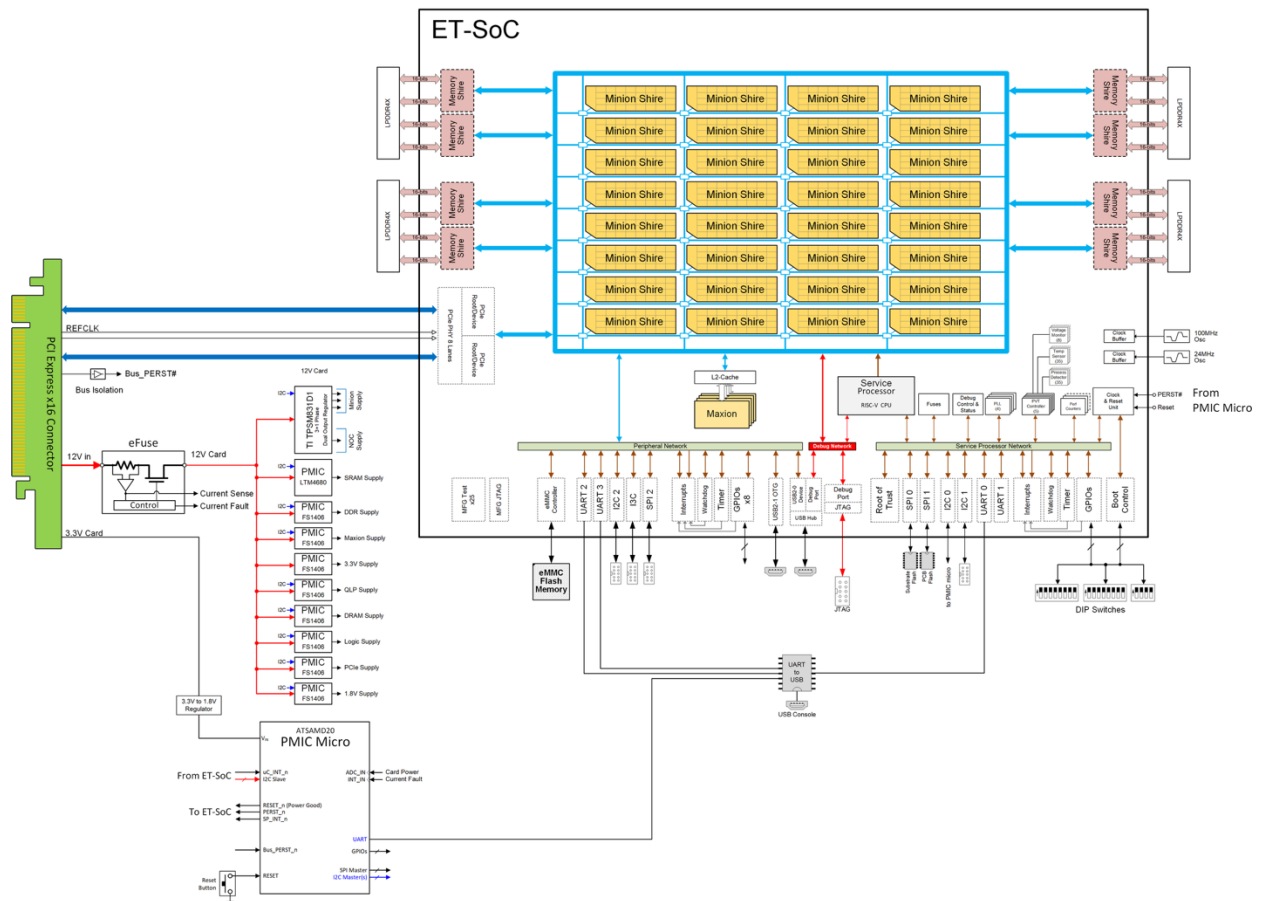


Features

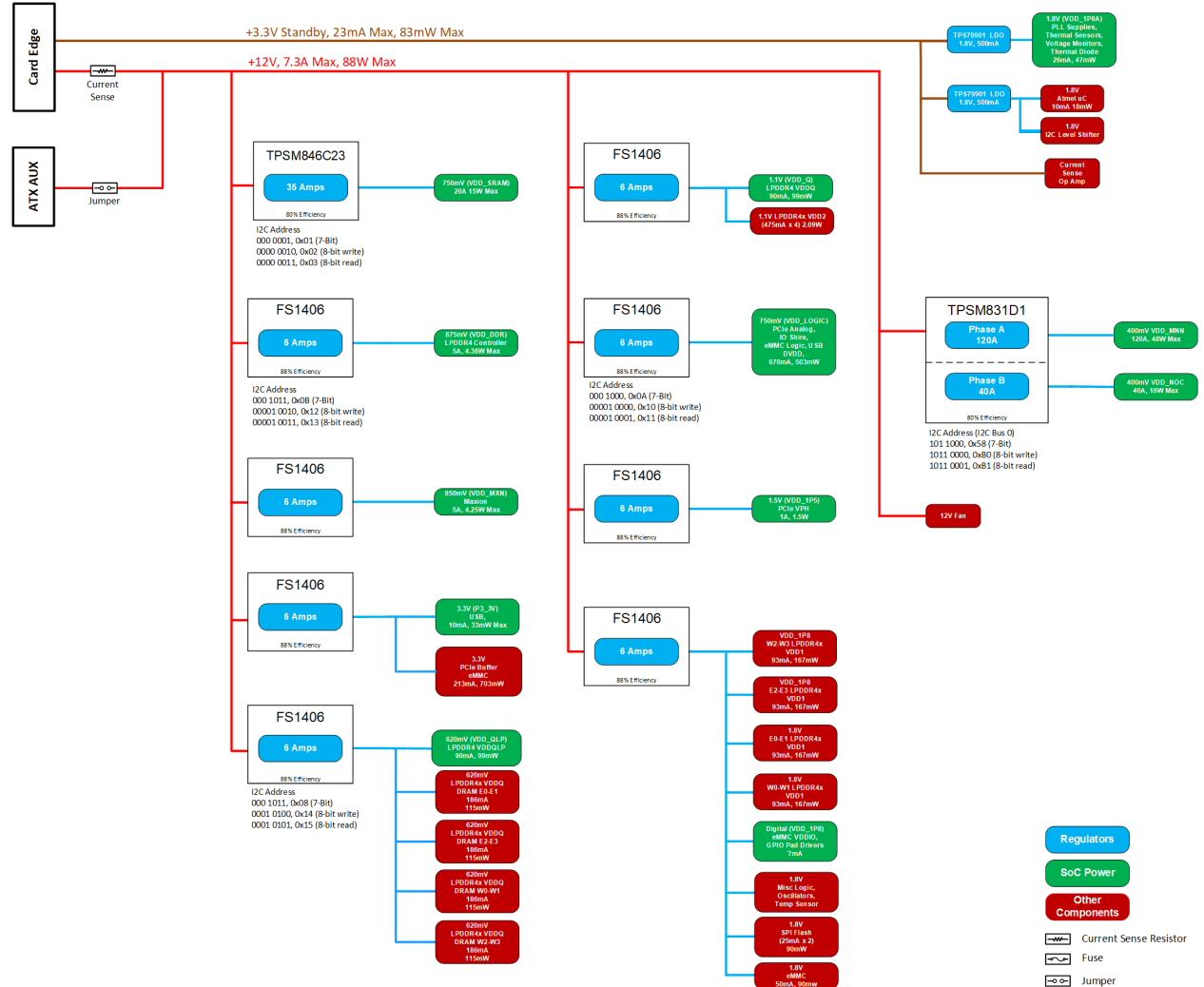
- PCIe x16 add-in card.
- PCIe x8 Gen 4 interface (x16 card edge connector).
- ET-SoC soldered-down processor (no socket option available).
- 4 x LPDDR4x DRAM devices for a combined 256-bit memory interface.
- USB type B connector for access to ET-SoC device port and UltraSOC debug port.
- USB type B micro connector for access to the FTDI multi-channel UART to USB interface.
- USB OTG micro-AB connector.
- JTAG connector for access to the ET-SoC JTAG interface.
- 64 GB eMMC flash memory device.
- Method for measuring total card power.
- DIP switches to set all boot options.

Block Diagram

PCIe V3 Development Card



Power Tree



(click to enlarge)

Key Voltage Regulators

Minion & NOC

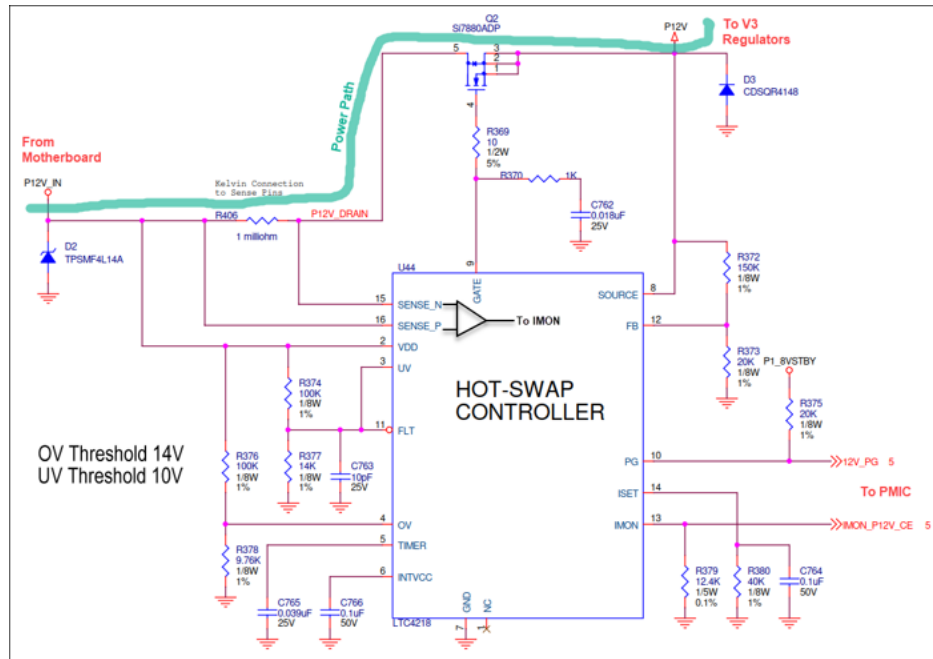
The Minion and NOC use the same Texas Instruments [TPSM831D31](#) voltage regulator module as BUB and PCIe V2 boards. This regulator operates as two independent voltage regulators that share a common PMBus controller, the NOC is connected to a single phase output with a maximum of 40A and the Minion connects to a 3-phase output with a maximum of 120A.

SRAM

The SRAM, aka: SCW voltage regulator is a Linear Technologies [LTM4680](#) PMBus voltage regulator module with a maximum of 60A.

Power Measurement

The V3 card contains a kelvin resistor on the 12V input to the card for measurement of total card power. This kelvin resistor is connects to a Linear Tech [LTC4218](#) "Hot Swap Controller". This controller is connected to a FET which turns on/off the 12V input to the V3 card.



(click to enlarge)

The LTC4218 performs the following functions:

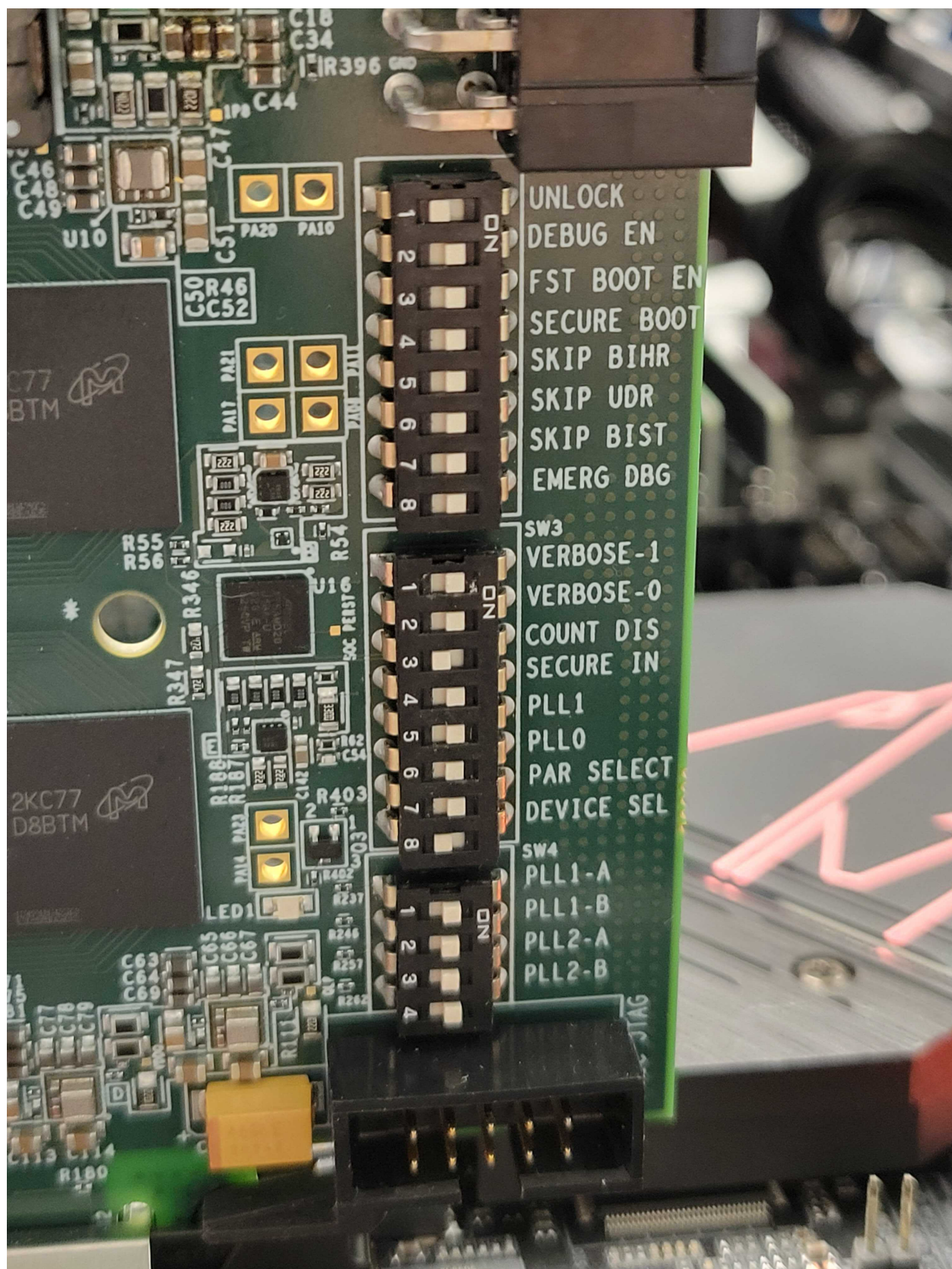
- **Over-voltage, under-voltage** control. The controller will only turn on power to the card when the input voltage is between 10 and 14V DC. This prevents the board from trying to turn on when the input voltage is not stable or will turn off power to the card if the input voltage goes too high.
- **Power good** monitoring. Provide a logic output indicating that the input voltage is within the 10 - 14V range.
- 5% accurate **current limit** with current foldback limiting. Using external components, a current limit is set and the controller will turn off power to the card if the current exceeds a set threshold.
- **Current monitor** output that amplifies the sense voltage for ground referenced current sensing. An analog output produces a current which is proportional to the current sensed across the kelvin resistor. The gain of the current sense amplifier is 100uA for 15mV on the kelvin resistor. In the V3 card the IMON output pin has a 12.4K 1% resistor to ground. This equals 1.24V on the IMON output pin when the card has 15mV across the

0.001 ohm kelvin resistor which would be 15A input current at 12V. This current would never be achieved and represents the top operating threshold of the current sense circuit.

DIP Switches

The following settings are used for screening and shipping to customers. Minor changes may need to be made in lab setting or DV settings but they must be set to the following in all the systems for running SW and shipping to customers.

V3.0 cards (301-325)



V3.2 cards (326-)

